WORKED SOLUTION

STEP I 1998, Question 2

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q2

The key idea

The awkward expression $(x^2+1)^{1/2}+x$ sits inside f, so the natural move is to call the whole thing t. The surprise is that this is easy to invert: squaring after isolating the root kills the x^2 terms and leaves x as a simple function of t. Once the first identity is in hand, the second integral is just a special case with $\mathrm{f}(t)=t^{-3}$.

Solution

The change of variable

Put

$$t=(x^2+1)^{1/2}+x.$$

As x runs from 0 to ∞, both $(x^2+1)^{1/2}$ and x are increasing, so t is strictly increasing, starting at $t=1$ when $x=0$ and tending to ∞ as $x\to\infty$. So the substitution is a one-to-one map of $[0,\infty)$ onto $[1,\infty)$, and the new limits are 1 and ∞.

? **Ask yourself** How do we get x back in terms of t?

Isolate the square root and square: $t-x=(x^2+1)^{1/2}$ gives

$$t^2-2tx+x^2=x^2+1 \quad\Longrightarrow\quad x=\frac{t^2-1}{2t}=\frac{1}{2}(t-t^{-1}),$$

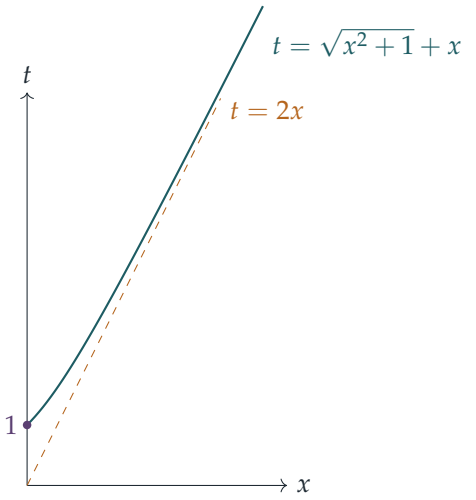
where dividing by t is fine because $t\geq 1$. (Squaring can introduce spurious solutions, but here $t-x=(t+t^{-1})/2>0$, so the root we squared really is positive.) Differentiating,

$$\frac{\mathrm{d}x}{\mathrm{d}t}=\frac{1}{2}(1+t^{-2}).$$

Therefore

$$\int_0^\infty \mathrm{f}((x^2+1)^{1/2}+x)\,\mathrm{d}x=\int_1^\infty \mathrm{f}(t)\cdot\frac{1}{2}(1+t^{-2})\,\mathrm{d}t=\frac{1}{2}\int_1^\infty(1+t^{-2})\mathrm{f}(t)\,\mathrm{d}t,$$

which is the required identity.



The sketch shows t rising steadily from 1, so no part of the range is covered twice.

The final part

Take $f(t) = t^{-3}$. The left-hand side of the identity becomes exactly the integral we want, so

$$\begin{aligned}\int_0^\infty ((x^2+1)^{1/2} + x)^{-3}\,\mathrm{d}x &= \frac{1}{2}\int_1^\infty (t^{-3} + t^{-5})\,\mathrm{d}t \\ &= \frac{1}{2}\left[-\tfrac{1}{2}t^{-2} - \tfrac{1}{4}t^{-4}\right]_1^\infty \\ &= \frac{1}{2}\left(\frac{1}{2} + \frac{1}{4}\right) = \frac{3}{8},\end{aligned}$$

as required. Both improper integrals converge, since t^{-3} and t^{-5} are integrable on $[1, \infty)$.

Ask yourself Is there a cleaner way to see the substitution?

Yes: put $x = \sinh\theta$. Then $(x^2+1)^{1/2} = \cosh\theta$, so $t = \cosh\theta + \sinh\theta = \mathrm{e}^\theta$. Our formula $x = \frac{1}{2}(t - t^{-1})$ is just $\sinh\theta = \frac{1}{2}(\mathrm{e}^\theta - \mathrm{e}^{-\theta})$ in disguise.

Checks

- At $x = 0$, $t = 1$, and $x = \frac{1}{2}(1-1) = 0$: the inverse formula agrees. At $x = \frac{3}{4}$, $t = \frac{5}{4} + \frac{3}{4} = 2$ and $\frac{1}{2}(2 - \frac{1}{2}) = \frac{3}{4}$.
- For large x, $t \approx 2x$, so the integrand in the final part behaves like $(2x)^{-3}$ and the integral certainly converges.

- Numerically, Simpson's rule on the original integral gives 0.375000, matching $\frac{3}{8}$. The identity in the first part also checks numerically for $\mathrm{f}(t) = \mathrm{e}^{-t}$ (both sides 0.258187) and $\mathrm{f}(t) = 1/(1+t^4)$ (both sides 0.188387).

Common slips

- Forgetting to change the limits: the lower limit becomes $t = 1$, not $t = 0$.
- Writing $\mathrm{d}t$ in terms of x and then struggling to eliminate x; it is far easier to express x in terms of t first and differentiate that.

Beyond STEP

The identity is a statement about changing variables under a monotonic map, which in analysis is justified for improper integrals by taking limits of integrals over $[0, X]$. The substitution itself is the inverse of sinh: $\ln\left(x + \sqrt{x^2+1}\right) = \operatorname{arsinh} x$, which is why t turns out to be $\mathrm{e}^{\operatorname{arsinh} x}$. For a careful treatment of improper integrals and substitution, see Bartle and Sherbert, *Introduction to Real Analysis*.

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